Computer Vision — 21CSE454T

Unit 4: Vision & Motion

Topic 2: Projection Schemes | SLO2

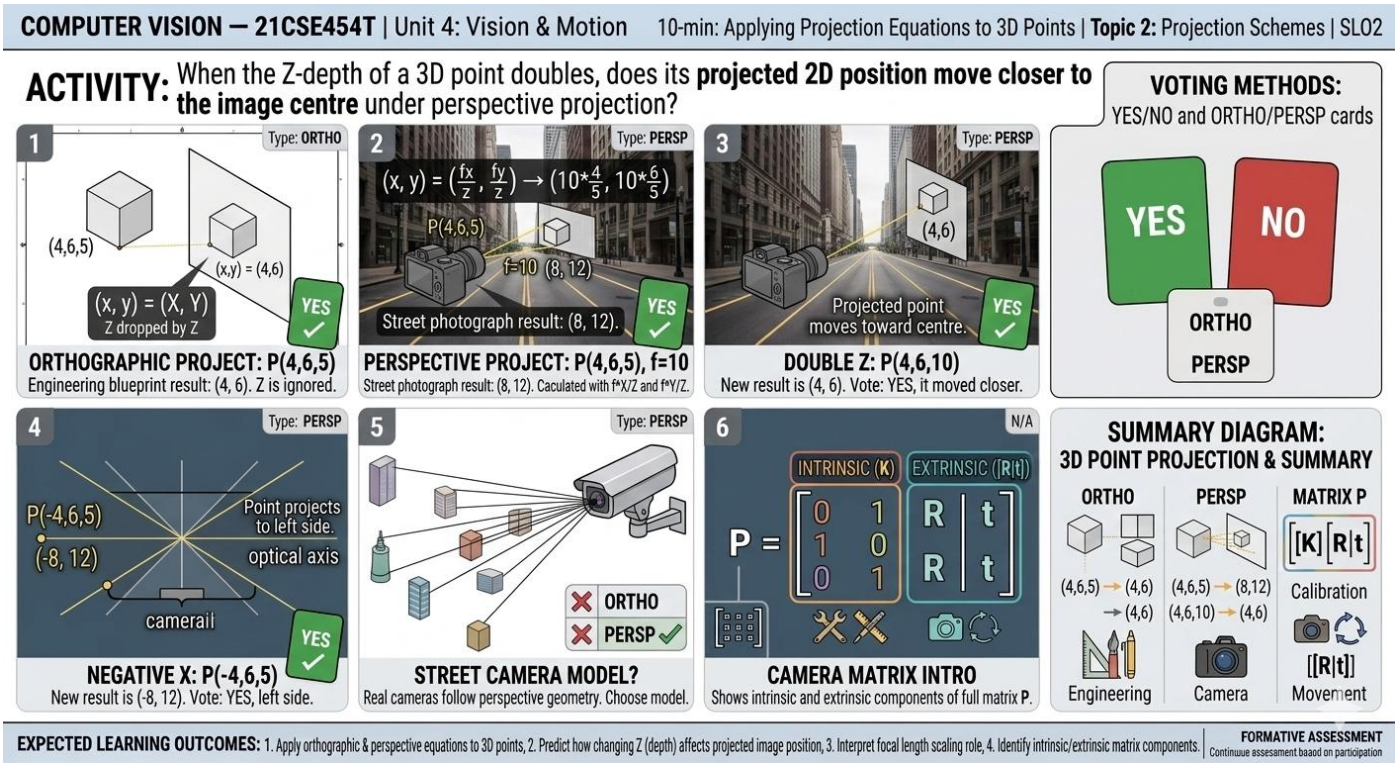
Course	Computer Vision — 21CSE454T
Unit	Unit 4: Vision & Motion
Topic	Topic 2: Projection Schemes
SLO2	Apply appropriate projection models to convert 3D objects to 2D representations.
Session Title	Applying Projection Equations to 3D Points
Duration	10 Minutes Activity
Response Method	YES / NO cards | Thumbs 👍 / 👎 | Labelled flashcards

Activity Question

> ***"When the Z-depth of a 3D point doubles, does its projected 2D position move closer to the image centre under perspective projection?"***

Activity Overview

Students trace how specific 3D points transform under orthographic and perspective projection equations, voting on predicted outcomes before the instructor reveals the calculation — building intuition for 3D-to-2D mapping.



Instructor Instructions

1. Write a 3D point P(4, 6, 5) on the board. Set focal length f = 10.
2. Walk through orthographic projection: (x, y) → (4, 6) — Z dropped.
3. Walk through perspective projection: (fx/z, fy/z) → (8, 12).
4. Change Z to 10 and ask: does the projected point move toward the centre? Students vote YES/NO.
5. Repeat with a negative X value and ask: which side of the image does the point project to?
6. Show the camera projection matrix [K][R|t] and explain each component.

Step-by-Step Activity Flow

Step	Action	Details / Expected Student Response
1	**Orthographic: P(4,6,5)**	Result is (4, 6). Students verify by dropping Z. Expected: YES, Z is ignored.
2	**Perspective: P(4,6,5), f=10**	Result is (8, 12). Students compute f*x/z and f*y/z. Check: YES, correct.
3	**Double Z: P(4,6,10)**	Perspective result is now (4, 6) — point moved toward centre. Vote: YES it moved closer.
4	**Negative X: P(-4,6,5)**	Perspective result is (-8, 12) — point projects to left side. Vote: YES, left side.
5	**Which model for street camera?**	Students vote ORTHO or PERSP. Answer: PERSP — real cameras follow perspective geometry.
6	**Camera matrix intro**	Show 3×4 projection matrix. Identify intrinsic (K) and extrinsic ([R|t]) parts.

Student Response Method

After each image or scenario is displayed, students respond using one of:

- Raising YES card (green) or NO card (red)
- Showing thumbs up 👍 for YES or thumbs down 👎 for NO
- Holding up labelled category cards (e.g., ORTHO / PERSP / OBLIQUE)

The instructor acknowledges the class response, reveals the correct answer, and provides a 1–2 sentence explanation before moving to the next item.

Expected Learning Outcomes

#	Learning Outcome
1	Apply orthographic and perspective projection equations to given 3D points.
2	Predict how changing depth (Z) affects a point's projected image position.
3	Interpret the role of focal length in perspective projection scaling.
4	Identify intrinsic and extrinsic components in the camera projection matrix.